Impella (Abiomed)

Guideline Title: **Impella (Abiomed)**	Guideline Number: **GUID-3203-PR030**

Issue date: TBA	Replaces Dept. Policy:
Revision dates: 4/28/22	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by: Ty Campbell
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To provide emergent and ongoing medical care and facilitate the safe transport of patients supported with Impella Catheters and an Automated Impella Controller (AIC).

II. **ABOUT THE IMPELLA SYSTEM:** The AIC and Impella Catheters have been cleared by the FDA for hospital-to-hospital transport via ambulance, helicopter, or fixed-wing aircraft. The Impella 2.5, 5.0, CP, and LD are intravascular microaxial blood pumps that support a patient's circulatory system. The Impella Catheter is inserted through the femoral or axillary artery into the left ventricle or, in the case of the LD, surgically via the ascending aorta. When properly positioned, the Impella delivers blood from the inlet area, which sits inside the left ventricle, through the cannula, to the outlet opening in the ascending aorta. Users monitor the correct positioning and functioning of the Impella on the display screen of the AIC. The pump allows the heart to rest and recover by off-loading the blood volume in the left ventricle into the ascending aorta. The pump improves cardiac output by improving stroke volume.

****The Abiomed Clinical Support Emergency Hotline is available 24/7 for technical assistance or troubleshooting at 1-800-422-8666**. There is also an Abiomed phone app for reference.**

III. **DEFINITIONS:**

a. ACT – Activated clotting time
b. AIC – Automated Impella Controller
c. CO – Cardiac output
d. CVP – Central Venous Pressure
e. MAP – Mean arterial pressure

IV. **EQUIPMENT**

a. Equipment for monitoring of heart rate, pulse oximetry, blood pressure, and EKG.
b. Automated Impella Controller Console.

c. Purge cassette and purge fluid.
d. Patient connector cable.
e. Impella securing straps for the aircraft

V. **DEPARTMENT GUIDELINE:**

a. Air Care Dispatch should make the medical crews aware of the transport request with this device prior to the transport.
b. Obtain an updated report from the sending facility medical team.
    i. Impella device type (2.5 liters, 5 liters, Impella LD, or Impella CP)
    ii. How catheter placement was confirmed.
    iii. Labs: CBC, BMP, Mg, BNP, aPTT, ACT and Chest Xray results
c. Establish care as defined by *GUID-G002 – General Patient Care*.
d. Complete a physical exam of patient to include
    i. Assess the catheter access site and dressing
        1. Assess securement of the catheter
        2. Maintain proper angle of entry of the catheter w/ 4x4s if needed
    ii. Assess urinary output and color. Report pink tinged or red color to physician and evaluate position of catheter and volume status (Target CVP > 10 mmHg if frequent suction alarms)
    iii. Assess distal perfusion of limb with Impella catheter – distal pulses & neurovascular check
    iv. Assess immobilization device if utilized.
    v. Head of bed must remain < 30 degrees. Pt may be turned carefully side to side.
e. Assess the system components
    i. Automated Impella Controller.
    ii. AIC batteries last 60 minutes when fully charged.
    iii. Purge cassette with tubing.
    iv. Patient connector cable (white cable with blue plug-in, disposable 1 per patient use).
    v. Purge fluid: At Orlando Health, the default purge solution is Heparin 12,500 units in 500 ml of D5W (25 units/ml). Dextrose Solution (Per physician's order, can be Dextrose 5-40% and may or may not contain Heparin. Never use saline solutions. This will damage the Impella system.
        1. Remove all air from bag to prevent air alarms in transport.
        2. Ensure there is enough purge solution to last for duration of the transport.
        3. If not, ask the sending providers to change the bag before transport.
        4. Solution is connected via the check valve side arm of the Impella on a Braun Vista Pump only. Rate is usually initiated at 7 ml/h with range of 4-12 ml/h.
    vi. Confirm Tuohy-Borst valve is locked (behind blue securement) and chart the CM marking on the Impella catheter closest to the Tuohy-Borst.

    vii. Note catheter depth closest to sheath. There is a numeric value every 5 cm and every marking in between is equal to 1 cm.
    viii. Confirm external fixations are secured.

f. Assess for normal operating conditions. If conditions not within range, consult with sending or receiving physician or Impella representative.
    i. Evaluate catheter position and monitor current waveform from the AIC placement screen. Proper placement is verified by an arterial waveform in the placement signal screen and a pulsatile motor current waveform.
    ii. Performance levels: P0 - P9 (should never be on P0 on a patient: causes retrograde flow, lowest on patient should be P2).
    iii. Impella flow: Based on 2.5 or 5.0 and performance level set. Maximum blood flow on 2.5 is 2.5 L/min and 5.0 maximum is 5 L/min.
    iv. Purge Pressures: 300 - 700 mmHg.
    v. Purge fluid flow: 4-12 mL/hr.
    vi. ACT: > 250 seconds.

g. Assess for Alarms and address alarms immediately.
    i. Critical Alarms: RED.
    ii. Serious Alarms: YELLOW.
    iii. Advisory Alarms: WHITE
    iv. Call support with any questions.
    v. Follow instructions on the AIC for troubleshooting alarms.
    vi. See Appendix for advanced troubleshooting
    vii. If air is detected in the system, press PURGE SYSTEM. Select “De-air pure system”. Follow on-screen prompts, making sure the bag is not inverted or empty. See Appendix.

h. Changing the purge system should NOT routinely need to be done by Air Care team members. If changing from a MPC (old console) to AIC (new console) be sure to obtain a new purge cassette and white patient specific cable. See appendix for reference of instructions.

i. Considerations prior to transport
    i. Confirm Impella placement with sending staff (echocardiography if needed).
    ii. The AIC should be fully charged prior to transport.
    iii. The AIC is designed to operate for at least 60 minutes on battery power. Crews must plug in the AIC into an AC power source in the aircraft during transports.
    iv. Keep the controller connected to AC power (or an AC inverter) whenever possible.
    v. Do not stress the connector cable from the controller to the Impella Catheter. Such tension could move the catheter out of correct position and compromise patient circulatory support.
    vi. The controller should be positioned to allow easy access to the display screen and soft buttons to view alarms and make any necessary changes.

vii. Reference: ACTs are typically maintained between 160-180 seconds or at the level ordered by the responsible physician. Review last patient ACT from sending facility staff.
viii. Do not use alcohol or alcohol-based cleaning solutions on the Impella purge sidearm.
ix. Contact the local team if the purge flow is < 3 ml/h

j. Prepare for transport with the Automated Impella Controller (AIC)
    i. Unplug the AIC from AC power.
    ii. Rotate the T-knobs to unlock the controller from the cart.
    iii. Lift the controller off the cart using the blue handle.
    iv. Place the controller on a flat surface (seat, bench, shelf, or stretcher) or hang the controller on the stretcher.
    v. Remove the purge solution from the IV pole on the cart and place it on the stretcher.
    vi. Secure the controller.
        1. If the controller is left unsecured, it can become a dangerous projectile.
        2. The Impella Controller has a bed mount on the back of the housing that may be used to hang the controller on the stretcher while the patient is transported inside the hospital.
        3. Straps may be used to secure the Impella Controller to a flat surface in the transport vehicle.
        4. The bed mount may be used as a loop through which to secure the straps

k. During care of the Impella Console
    i. Planning is critical. Contact **Abiomed Clinical Support if needed: 1-800-422-8666**.
    ii. Do not cover the pump up with sheet or blankets.
    iii. Maintaining optimal patient hemodynamic status and correct Impella position are two key factors in managing patients supported with the Impella System during transport. Steps should be taken to eliminate or minimize any aspect of the transport that might adversely affect these two factors.
    iv. Monitor patient hemodynamics, pump flow, and motor current. Carefully monitor purge pressures during changes in altitude.
        1. Use a MAP to guide therapy. Maintain MAP > 65 mmHg unless other goals ordered.
            a. Titrate inotropes and vasopressors to achieve this goal.
            b. Increase the P-level one level at a time to achieve MAP goals in transport.
            c. Infusions should be weaned before decreasing the P-level.
        2. Maintain a CVP of 10 mmHg.

a. If the Max and min flow are separated by more than 1 L/min, this can indicate the patient is dry and will likely respond to a fluid bolus.

3. Maintain recommended purge pressure estimated 300 – 700 mmHg.

v. Manage groin site
1. Head of bed must remain less than 30 degrees.
2. Keep leg with the pump insertion site straight. Use knee immobilizer if available.
3. Use aseptic technique if dressing change is needed.
4. Assess access site for bleeding and hematoma
5. Ensure device is anchored to the leg.
6. Be careful not to pull on the Impella Catheter when transferring a patient from one bed to another.
7. Monitor pedal pulses.

vi. If possible, place the controller and the red Impella plug at the level of the patient's heart during transport.

vii. If you suspect improper placement (aortic or ventricular) of Impella Catheter, drop to P2 performance level on AIC and treat patient's hemodynamics.

viii. Defibrillate when indicated. The Impella system does not have to be stopped or unplugged to defibrillate.

ix. If cardiac arrest occurs:
1. Start compressions when indicated.
2. Drop the performance level to P2 with CPR. (Press MENU tab – Flow Control – Turn grey dial to adjust P level – Push grey dial in to confirm change).
3. When return of spontaneous circulation, check Impella catheter placement. If it appears correct, slowly increase patient back to previous performance level. If placement cannot be verified, maintain performance level at P2 & treat medically.

x. If you need to de-air the purge system, see Appendix for instructions.

xi. If an Intra-aortic balloon pump is in place, augmentation should be set at 50% and frequency should be set at 1:3.

xii. Narrow pulsatility: Your waveforms may be very dampened. The Impella may give a grey alert "Impella position unknown due to low native heart pulsatility". This is a reflection of the narrowed pulse pressure, not a position issue. If the MAP >60 mmHg, patient is perfusing.

l. Ongoing assessment and charting:

i. Peripheral pulses.

ii. Insertion site: Femoral artery, or axillary artery. The 5.0 will be a cut down direct in aorta

iii. Monitor pump placement using the dual signal waveforms of the motor current & placement signal.
iv. Placement signal values
    1. On 2.5 or Impella CP: not true arterial pressure reading but can be used as a trend guide. The placement signal should display an aortic tracing with the open pressure port in the ascending aorta.
    2. On 5.0 and Impella LD: this will be a differential pressure number (may look like a left ventricular tracing). If there is no pulsatility: catheter is out of place or patient has little native LV function left.
    3. Do not use the arterial pressure on the console as a monitor of the patient's blood pressure. Note: the RED placement signal AO (aortic pressure) is an estimate pressure as there is no transducer. It is a placement signal only, not a true pressure. True blood pressure must be taken with another a-line or with cuff. The mean AO pressure represents afterload and goal is 60-80 mmHg. The AO pressure waveform should always be pulsatile as should every aortic waveform. The AO peak should always be in line with the left ventricular pressure waveform. This will demonstrate correct placement. There will be no LV waveform present with a P-level 3 or less.
v. The GREEN motor current value and waveform should always look pulsatile like an aortic waveform. A flat motor waveform can indicate that the device is entirely in the left ventricle or entirely in the aorta and device will need to be repositioned.
vi. If the catheter migrates to aortic placement or ventricular placement, do NOT attempt to reposition. Immediately turn the P level to 2 and treat the patient medically as the Impella is no longer properly placed. Notify the receiving unit or local representative ASAP.
vii. Maintain pressurized flush bag of normal saline to the red sidearm of the Impella device.
viii. Flow rate: The Impella flow shows the maximum systolic and maximum diastolic Impella assisted CO in liters/min. This is the CO only being provided by the Impella.
    1. Impella 2.5 (flow rates up to 2.5 LPM)
    2. Impella 5.0 (flow rates up to 5 LPM)
    3. Impella CP (flow rates up to 3.5 LPM)
    4. Impella LD (up to 5 LPM, surgically placed into aorta through open chest surgery)
    5. Impella RP (right ventricular)
ix. P-level: Indicates the power/performance level of the motor while running from P9 (maximum) to P0. Do not decrease below P2 as long as the pump is in the ventricle.

VI. **DOCUMENTATION:** EMS Charts

a. Documentation should include: Impella size, motor current (A), Speed (RPM), pump position, placement signal (mmHg), Performance (P) level, Impella flow rate (L/min) (minimum, maximum, & mean), purge fluid flow/infusion rate (ml/h), purge pressure (mmHg), purge fluid type & concentration, insertion site placement/condition, insertion length, distal pulses, most recent ACT.
b. Documentation every 15 minutes: P-Level, Flow rate (L/min), & satisfactory placement signal and motor current.
c. Document each intervention and change in P-level during care

VII. **REFERENCES:**

a. ABIOMED. (2011, December). Patient Transport with the Automated Impella Controller.
b. ABIOMED. (2020). Video: OR and ICU Checklist Best Practices for Impella 5.5 with SmartAssist.
c. ProtectedPCI.com. (2020). Best practices supported with Impella (part 5): Impella groin management.
d. ProtectedPCI.com. (2020). Best practices supported with Impella (part 4): Addressing Impella alarms.
e. ABIOMED. (2020). De-Airing the Impella purge System.
f. Orlando Health: Impella Ventricular Assist Device Post Insertion Orders
g. ORMC ICU Training and Reference Manual

VIII. **APPENDIX**

a. **Impella Transfer Directions** per ABIOMED 10/28/2010
   i. **How to transfer support from the Impella Console to the new Impella Controller (AIC)**
      1. Prepare
         a. Turn on the Impella Controller.
         b. Open the purge cassette and spike a new bag of purge solution.
         c. Insert the purge cassette into the controller.
         d. Open the white connector cable and plug into the controller.
      2. Switch
         a. Disconnect the blue cable from the catheter.
         b. Connect the white cable to the catheter.
         c. Confirm re-starting the catheter on the controller. (The MPC system will still be purging the catheter at this point.)
      3. Prime
         a. Press PURGE SYSTEM on the controller and select Change Purge Fluid. NOTE: There may be purge alarms on the console. Continue with the purge fluid change to complete the transfer. Check the alarms once the transfer has been completed.

b. Complete the purge fluid change making sure to flush the purge tubing.
c. Disconnect old purge tubing.
d. Connect new purge tubing.

4. Address any unresolved alarms by following the help text displayed.
5. Tighten the Tuohy-Borst valve on the Impella Catheter to prevent catheter migration. (Tighten all the way to the right).
6. For patients supported with the Impella 2.5 Catheter, attach a saline pressure bag pressurized between 300-350 mmHg to the red sidearm and complete the "Transfer to Standard Configuration" under the PURGE SYSTEM menu.

ii. **How to transfer support from a new Automated Impella Controller to an Impella Console**

1. Gather the following items prior to switching support: Impella Console system (Console, infusion pump, and Impella Power Supply on cart), blue connector cable, grey pressure transducer cable, infusion pump tubing, CM-Set, purge solution.
2. Prepare
    a. Turn on the Impella Controller.
    b. Setup the infusion pump and tubing.
    c. Connect the grey pressure cable to the transducer and the console.
    d. Plug the blue cable into console.
    e. Press PURGE SYSTEM on the Automated controller and select Change Purge Fluid on the Impella Controller to complete the procedure to bolus the purge system (Do not flush the purge fluid from the cassette).
3. Switch
    a. Disconnect the purge cassette tubing (Alarm will sound PURGE SYSTEM OPEN).
    b. Connect the new purge tubing and start the infusion.
    c. Disconnect the white cable from the Impella Catheter.
    d. Connect the blue cable to the Impella Catheter.
4. Tighten the Tuohy-Borst valve on the Impella Catheter to prevent catheter migration. (Tighten all the way to the right)
5. For patients supported with the Impella 2.5 Catheter, attach a saline pressure bag pressurized between 300-350 mmHg to the red sidearm.

iii. **Transferring support from the Automated Impella controller to a new Automated Impella Controller**

1. Gather the AIC
2. Prepare

            a. Confirm that the new controller is powered on and ready.
            b. Press PURGE SYSTEM on the original controller, select Change Purge Fluid, and complete the procedure to bolus the purge system. (Do NOT flush the purge fluid from the cassette).
            c. Disconnect the yellow luer connector from the Impella Catheter to release the pressure in the purge cassette.
        3. Switch
            a. Transfer the purge cassette and purge solution from the original controller to the new controller.
            b. Reconnect the yellow luer connector to the Impella Catheter.
            c. Remove the white connector cable from the original controller and plug it into the catheter plug on the front of the new controller.
        4. Confirm
            a. Once the Impella Catheter is connected to the new controller, a message will appear on the screen asking you to confirm re-starting the Impella Catheter at the previously set flow rate.
            b. Press OK within 10 seconds to confirm restarting the Impella Catheter
        5. Tighten the Tuohy-Borst Valve on the Impella Catheter to prevent catheter migration. (Tighten all the way to the right)
        6. For patients supported with the Impella 2.5 Catheter, attach a saline pressure bag pressurized between 300-350 mmHg to the red sidearm and complete the “Transfer to Standard Configuration” under the PURGE SYSTEM menu.
b. **Improved Troubleshooting for Alarm Management**
    i. Suction Alarms
        1. Causes: Inadequate LV filling, hypovolemia, Incorrect position, RV failure.
        2. Effects of suction: Lower Impella flow, decreased benefit of device, risk of hemolysis, lower blood pressure
        3. Diastolic Suction
            a. Signs: Negative diastolic pressure and recovers by end of diastole, normal systolic pressure, or low diastolic flows
            b. Resolve: Reduce P level by 2. Check filling and volume status and Impella position.
                i. If CVP < 10mmHg, administer volume
        4. Continuous suction

        a. Signs: Negative diastolic pressure and does not recover, low systolic pressure, & low systolic & diastolic flows
        b. Resolve: Reduce P level by 2. Check Impella position then filling and volume status
    5. If the suction alarm resolves, attempt to resume the previous P level
    6. If the suction alarm continues administer additional volume.
ii. Purge System Blocked Checklist
    1. Purge Flow <2ml/h and Purge pressure > 700 mmHg
        a. Asses the tubing:
            i. Are there any kinks in the purge tubing the clear sidearm, or anywhere along the catheter? Make sure the roller clamp on saline bag is not closed. Ensure pressure bag is not inflated > 350 mmHg.
            ii. Respond: Straighten the tubing, clear the sidearm, or catheter
        b. Assess the purge fluid
            i. Is the purge fluid concentration too high?
            ii. Respond: Consider discussion with the physician to reduce the purge fluid (dextrose) concentration
        c. Assess the Motor Current
            i. If unable to resolve purge system blocked alarm, monitor for increases in motor current which can indicate impending pump failure
            ii. Call physician. May need tPA or to replace the pump.
iii. Low Pulsatility: Patients with profound left ventricular dysfunction may have low native heart pulsatility. Decreased pulsatility, which is common and represents a low flow state in the ventricle, can trigger an Impella Position Unknown alarm.
    1. Alarm: "Impella position unknown due to low pulsatility, assess cardiac function
    2. Assess
        a. Check placement signal: signal pule pressure narrowed to < 20 mmHg
        b. Motor current: may be pulsatile but dampened
    3. Treat
        a. Consider increasing Impella flow P level
        b. Assess position via echo when at facility
        c. Treat the patient based on MAP
            i. MAP > 60 mmHg: no action needed
            ii. MAP < 60 mmHg: Escalate therapy and consider inotropes

        iii. Treat the patient, not the device

c. **De-Airing the Impella Purge System**
    i. Select the purge system menu soft button
    ii. Use the selector knob to go to ‘de-air purge system’ and press in on the selector knob.
    iii. Press start to begin the procedure
    iv. Ensure the purge fluid line is not kinked and the bag is not inverted. Once verified click NEXT.
    v. Disconnect the yellow luer from the Impella catheter. Do not use a hemostat to disconnect the luer as this could as this could damage the Impella catheter.
    vi. The controller will begin to prime the tubing. Do not connect the luer during priming.
    vii. Confirm there is no air in the tubing. Then press next.
    viii. Connect the yellow luers together in a wet-to-wet connection.
    ix. Press done to exit the purge menu.

Purge Pressure	**Purse Flow Rate**	**Additional Action**	**Dose Change of the Impella Heparin Purse Solution**
< 300 mmHg	< 12 ml/h	Bolus 1 ml of the Impella Heparin Purge Solution	-Increase infusion rate by 1ml/h to maintain purse pressure > 300 mmHg -If low pressure alarms remain unresolved for more than 5 minutes, notify MD
< 300 mmHg	> 12 ml/h	Tighten loose connections. Bolus 1 ml of the Impella Heparin Purge Solution & Notify MD	-Per MD and Abiomed Clinical Support to increase concentration. -If alarms remain unresolved for 5 minutes despite action consider pump damage.
300-700 mmHg (recommended range)	4 – 12 ml/h	None	None
> 700 mmHg	< 4 ml/h	Inspect tubing for kinks or closed clamps & notify MD	Per MD and Abiomed Clinical Support
> 700 mmHg	> 4 ml/h	Inspect tubing for kinks or closed clamps	-Decrease infusion rate by 1 ml/h -If high pressure alarms remain unresolved for > 2 hours, notify MD.

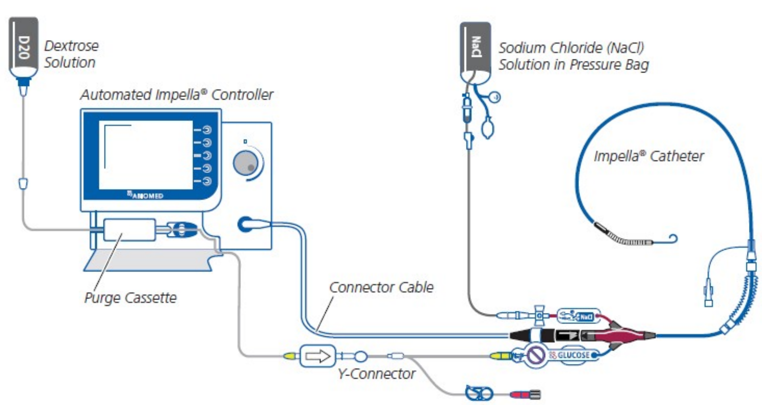

Figure 3.3 Standard Configuration of the Automated Impella® Controller, Impella® Catheter, and Accessories